



National
Qualifications
2015

2015 Mathematics

National 5 Paper 1

Finalised Marking Instructions

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Detailed Marking Instructions for each question

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
1.			Ans: $3\frac{13}{15}$ or $\frac{58}{15}$ • ¹ correct common denominator • ² correct answer	2	• ¹ e.g. $6\frac{3}{15} - 2\frac{5}{15}$ or $\frac{93}{15} - \frac{35}{15}$ • ² $3\frac{13}{15}$ or $\frac{58}{15}$

Notes:

- Correct answer without working award 0/2
- Do not penalise incorrect conversion of $\frac{58}{15}$ to a mixed number

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
2.			Ans: $x > -5$ • ¹ multiply out bracket • ² collect like terms • ³ solve for x	3	• ¹ $11 - 2 - 6x < 39$ • ² $-6x < 30$ or $-30 < 6x$ • ³ $x > -5$ or $-5 < x$

Notes:

- Correct answer without working award 1/3
- (a) For $11 - 2 - 6x < 39 \rightarrow 6x < 30 \rightarrow x < 5$ award 1/3 ✓ x x
 (b) For $11 - 2 + 6x < 39 \rightarrow 6x < 30 \rightarrow x < 5$ award 1/3 x ✓ x
- For $9(1 + 3x) < 39 \rightarrow 9 + 27x < 39 \rightarrow 27x < 30 \rightarrow x < \frac{30}{27}$ award 1/3 x ✓ x

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
3.			Ans: 39° • ¹ calculate the size of angle OBD • ² calculate the size of angle EDF • ³ calculate the size of angle BDF	3	• ¹ angle OBD = 13° • ² angle EDF = 26° • ³ angle BDF = 39°
Notes: 1. The first two marks may be awarded for information marked on the diagram 2. An answer of 39° must be stated outwith the diagram for the third mark to be awarded 3. Third mark is only available where angle ODB = angle OBD 4. For an answer of 39° with no relevant working award 0/3					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
4.			Ans: $x^3 - 3x^2 - 6x + 8$ • ¹ start to multiply out brackets • ² complete multiplying out brackets • ³ collect like terms which must include a term in x^3	3	• ¹ evidence of 3 correct terms eg $x^3 + x^2 - 2x$ • ² $x^3 + x^2 - 2x - 4x^2 - 4x + 8$ • ³ $x^3 - 3x^2 - 6x + 8$
Notes: 1. Correct answer with no working award 3/3					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
5.			Ans: $a = 8$ •¹ find $\bar{x}$ and $(x - \bar{x})^2$ •² substitute into formula for a •³ calculate value of a	3	•¹ 3 and 4, 1, 1, 1, 25 •² $\frac{32}{5-1}$ •³ 8

Notes:

1. Where a candidate has worked out the standard deviation award marks as follows:

•¹ find $\bar{x}$ and $(x - \bar{x})^2$

•¹ 3 and 4, 1, 1, 1, 25

•² substitute into formula

•² $\sqrt{\frac{32}{5-1}}$

•³ calculate standard deviation

•³ $\sqrt{8}$

2. For use of alternative formula award marks as follows:

•¹ find $\sum x$ and $\sum x^2$

•¹ 15 and 77

•² substitute into formula for a

•² $\frac{77 - \frac{15^2}{5}}{5-1}$

•³ calculate value of a

•³ 8

3. For a final answer of $a = \sqrt{8}$ award 2/3

4. Disregard any attempt to simplify $\sqrt{8}$

5. Correct answer without working award 0/3

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
6.			Ans: $a = 4, b = 3$ • ¹ state the value of a • ² state the value of b	2	• ¹ 4 • ² 3
Notes: 1. For an answer of $y = 4\sin 3x$ award 2/2 2. For an answer $a = 3, b = 4$ or $y = 3\sin 4x$ award 1/2					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
7.	(a)	(i)	Ans: -2 • ¹ state value of a	1	• ¹ -2
		(ii)	Ans: -4 • ¹ state value of b	1	• ¹ -4
Notes: 1. Where a candidate has answers of (i) -4 and (ii) -2 award 0/1 for (i) and 0/1 for (ii)					
	(b)		Ans: $x = 2$ • ¹ state equation of axis of symmetry	1	• ¹ $x = 2$
Notes: 1. For answers of 2 or axis of symmetry = 2 award 0/1					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
8.			Ans: $y = 2x + 9$ •¹ find gradient •² substitute gradient and a point into $y - b = m(x - a)$ or $y = mx + c$ •³ state equation of the line in terms of y and x in its simplest form.	3	•¹ $\frac{10}{5}$ •² e.g. $y - 15 = \frac{10}{5}(x - 3)$ or $15 = \frac{10}{5} \times 3 + c$ •³ $y = 2x + 9$
Notes: 1. Correct answer without working award 3/3 2. For a final answer of $y = \frac{2}{1}x + 9$ award 2/3 ✓✓x					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
9.			Ans: $\cos 100^\circ$, $\cos 90^\circ$, $\cos 300^\circ$; with justification •¹ state correct order •² justification stated explicitly	2	•¹ $\cos 100$, $\cos 90$, $\cos 300$ •² $\cos 100$ is negative, $\cos 90$ is zero and $\cos 300$ is positive (or similar)
Notes: 1. Where 2 out of the 3 values are in the correct position relative to each other, with valid reason award 1/2 e.g. For “$\cos 90^\circ$ is positive, $\cos 100^\circ$ is negative, $\cos 300^\circ$ is positive; so $\cos 100^\circ$, $\cos 300^\circ$, $\cos 90^\circ$” award 1/2 2. Accept positions of $\cos 90^\circ$, $\cos 100^\circ$ and $\cos 300^\circ$ indicated on a cosine curve for award of the second mark					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
10.	(a)		Ans: median = 19.5, SIQR = 4.5 • ¹ find median • ² find quartiles • ³ calculate semi-interquartile range	3	• ¹ 19.5 • ² 17 and 26 • ³ 4.5
Notes: 1. An incorrect answer for the median must be followed through with the possibility of awarding marks 2 and 3 2. If 'correct' SIQR is found from an (a) ordered list with one missing or one extra number award 2/3 x✓✓ (b) unordered list award 1/3 xx✓					
	(b)		Ans: valid comments • ¹ compare medians • ² compare semi-interquartile ranges	2	• ¹ On average the second round's scores are higher • ² The second round's scores are more consistent.
Notes: 1. Answers must be consistent with answer to part (a) 2. Statements must show understanding of the concepts e.g. (a) "In general the second round's scores were higher" is acceptable <u>but</u> "The median of the second round was higher" or "The second round's scores were higher" are not acceptable. (b) "The spread of scores in the second round was lower" is acceptable <u>but</u> "the range of scores in the second round was lower" is not acceptable.					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
11.			Ans: $x=7, y=-2$ • ¹ evidence of scaling • ² follow a valid strategy through to produce values x and y • ³ calculate correct values for x and y	3	• ¹ $6x+4y=34$ $6x+15y=12$ • ² values for x and y • ³ $x=7$ and $y=-2$
Notes: 1. For a solution obtained by guess and check award 0/3					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
12.			Ans: $\frac{x}{x+5}$ • ¹ factorise numerator • ² factorise denominator • ³ cancel brackets correctly	3	• ¹ $x(x-4)$ • ² $(x-4)(x+5)$ • ³ $\frac{x}{x+5}$
Notes: 1. Correct answer without working award 3/3 2. For subsequent incorrect working, the final mark is not available					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
13.			Ans: $\sqrt{2}$ • ¹ express as equivalent fraction with rational denominator • ² manipulate surds • ³ consistent answer	3	• ¹ $\frac{4\sqrt{8}}{8}$ • ² $\frac{4 \times 2\sqrt{2}}{8}$ • ³ $\sqrt{2}$

Notes:

1. Alternative strategy:

•¹ manipulate surds

•¹ $\frac{4}{2\sqrt{2}}$

•² express as equivalent fraction
with rational denominator

•² $\frac{4\sqrt{2}}{2 \times 2}$

•³ consistent answer

•³ $\sqrt{2}$

2. For an answer of $\frac{4\sqrt{8}}{8} \rightarrow \frac{\sqrt{8}}{2}$ award 1/3

3. Correct answer with no working award 0/3

4. All steps must be shown

e.g. For $\frac{4}{2\sqrt{2}} = \sqrt{2}$ with no intermediate steps shown award 1/3

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
14.			Ans: 32 • ¹ interpret index • ² complete evaluation	2	• ¹ $\sqrt[3]{8^5}$ • ² 32

Notes:

1. Correct answer without working award 2/2

2. For $\sqrt[3]{8} = 2$ or $8^5 = 32768$ award 1/2

[END OF MARKING INSTRUCTIONS]



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National 5 Paper 2

Finalised Marking Instructions

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Detailed Marking Instructions for each question

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
1.			Ans: £253 628 (·16) • ¹ know how to increase by 2·8% • ² know how to calculate expected turnover • ³ carry out calculations correctly within a valid strategy	3	• ¹ $\times 1\cdot028$ • ² $240\,000 \times 1\cdot028^2$ • ³ $253\,628 (\cdot16)$

Notes:

- For an answer of 253 628 without working award 3/3
- Where an incorrect percentage is used, the working must be followed through to give the possibility of awarding 2/3
e.g. for an answer of 393 216 ($240\,000 \times 1\cdot28^2$), with working award 2/3
- For an answer of 246 720 ($240\,000 \times 1\cdot028$), no working necessary award 1/3
- For an answer of 493 440 ($240\,000 \times 1\cdot028 \times 2$), with working award 1/3
- For an answer of 253 440 ($240\,000 + 240\,000 \times 0\cdot028 \times 2$), with working award 1/3
- For an answer of 13 440 ($240\,000 \times 0\cdot028 \times 2$) award 0/3

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
2.			Ans: $a=7$ • ¹ valid strategy • ² state value of a	2	• ¹ $3a+2=23$ or $3\times7+2(=23)$ • ² 7

Notes:

- Correct answer without working award 2/2
- Accept $x=7$ award 2/2
- For an answer of $3\times23+2=71$ award 0/2

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
3.			Ans: 0.78 km • ¹ correct substitution into cosine rule • ² evaluate AB^2 • ³ calculate AB	3	• ¹ $1.35^2 + 1.2^2 - 2 \times 1.35 \times 1.2 \times \cos 35^\circ$ • ² 0.608..... • ³ 0.78

Notes:

- For 0.8 with valid working award 3/3
- Disregard errors due to premature rounding provided there is evidence
e.g. $1.35^2 + 1.2^2 - 2 \times 1.35 \times 1.2 \times 0.8 = 0.6705 \Rightarrow \text{final answer} = 0.82$ award 3/3
- Correct answer without working award 0/3
- For 2.49 (uses RAD) or 0.71 (uses GRAD), with working award 3/3

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
4.			•Ans: 23 • ¹ start process • ² solution	2	• ¹ $6^2 + (-13)^2 + 18^2$ • ² 23

Notes:

- Correct answer without working award 2/2
- For 13.8.....(e.g. $\sqrt{6^2 - 13^2 + 18^2}$) , no working necessary, award 1/2

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
5.			Ans: $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$ • ¹ state components of either vector p or vector q • ² state components of vector p and vector q and vector p + q	2	• ¹ $\begin{pmatrix} -5 \\ 3 \end{pmatrix}$ or $\begin{pmatrix} 4 \\ -5 \end{pmatrix}$ • ² $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$

Notes:

1. Alternative method:

- ¹ correct nose to tail diagram (must include arrows)
- ² state components of vector **p + q**

2. Correct answer without working award 2/2

3. Special cases (working must be shown)

(a) $\begin{pmatrix} 5 \\ -3 \end{pmatrix} + \begin{pmatrix} -4 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ award 1/2 x✓

(b) $\begin{pmatrix} 3 \\ -5 \end{pmatrix} + \begin{pmatrix} -5 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$ award 1/2 x✓

4. For $(-1, -2)$ award 1/2

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
6.	(a)		Ans: $1.1 \times 10^{12} \text{ km}^3$ • ¹ substitute radius into volume of a sphere formula • ² evaluate volume • ³ round volume to 2 significant figures	3	• ¹ $V = \frac{4}{3} \times \pi \times (6400)^3$ • ² $1.098... \times 10^{12}$ • ³ 1.1×10^{12}
Notes: 1. Accept variations in π 2. Some answers (without working) (a) 1.1×10^{12} award 3/3 (b) 1.10×10^{12} (2 d.p.) award 2/3 ✓✓× (c) 1.0×10^{12} award 0/3 3. Some answers (working must be shown) (a) $\frac{4}{3} \times \pi \times (6400)^2 = 1.71..... \times 10^8 = 1.7 \times 10^8$ award 2/3 ×✓✓ (b) $\frac{4}{3} \times \pi \times 6400 = 2.68..... \times 10^4 = 2.7 \times 10^4$ award 1/3 ××✓					
	(b)		Ans: 50 times bigger • ¹ know to divide earth volume by moon volume • ² divide correctly	2	• ¹ $\frac{1.1 \times 10^{12}}{2.2 \times 10^{10}}$ • ² 50
Notes: 1. Correct answer without working award 2/2 2. $\frac{1.098..... \times 10^{12}}{2.2 \times 10^{10}} = 49.9....., 50 \text{ or } 49$ award 2/2					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
7.			Ans: 10s • ¹ know how to start division calculation • ² continue process • ³ express in simplest form	3	• ¹ $\frac{5t}{s} \times \frac{2s^2}{t}$ or equivalent • ² evidence of correctly cancelling either variable or $\frac{10ts^2}{st}$ • ³ 10s
Notes: 1. Correct answer without working award 3/3 2. For $\frac{10s}{1}$ award 2/3 ✓✓x					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
8.			Ans: £350 • ¹ know that 85% = £297·50 • ² begin valid strategy • ³ answer	3	• ¹ 85% = 297·50 • ² $1\% = \frac{297 \cdot 50}{85}$ (=3·5) • ³ $100\% = \frac{297 \cdot 50}{85} \times 100 = 350$
Notes: 1. For 350 with or without working award 3/3 2. For 252·88 (85% of 297·50) or 342·13 (115% of 297·50) (i) and evidence of 85% = 297·50 award 1/3 ✓xx (ii) otherwise award 0/3 3. For 115% = 297·50 → 258·70 award 2/3 xx✓✓ 4. For subsequent incorrect working, the final mark is not available e.g. 350 + 297·50 = 647·50 award 2/3 ✓✓x					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
9.			Ans: 225 cm^2 • ¹ linear scale factor • ² know how to find area of triangle PRS • ³ find area of triangle PRS • ⁴ find area of quadrilateral PQTS	4	• ¹ $\frac{30}{24}$ • ² $\left(\frac{30}{24}\right)^2 \times 400$ • ³ 625 • ⁴ 225

Notes:

- (a) $\frac{30}{24} \times 400 = 500$ award 1/4 ✓xxx
 (b) $\frac{30}{24} \times 400 - 400 = 100$ award 2/4 ✓xx✓
 (c) $\left(\frac{30}{24}\right)^3 \times 400 - 400 = 381 \times 25$ award 3/4 ✓x✓✓
- Where premature rounding leads to an inaccurate answer the third mark is not available
 e.g. $\frac{30}{24} = 1.25 \Rightarrow 1.3^2 \times 400 = 676 \rightarrow 276$ award 3/4 ✓✓x✓
- The fourth mark is not available where area of triangle PRS is less than 400
 e.g. $\left(\frac{24}{30}\right)^2 \times 400 = 256$ award 2/4 x✓✓x
- Where candidate assumes that triangles are right-angled the maximum available mark is 3/4 x✓✓✓ (but see note 2 above)
 - ² $QR = \frac{400}{\frac{1}{2} \times 24} = 33 \frac{1}{3} \rightarrow PR = \frac{30}{24} \times 33 \frac{1}{3} = 41 \frac{2}{3}$
 - ³ area of $PRS = \frac{1}{2} \times 41 \frac{2}{3} \times 30 = 625$
 - ⁴ area of $PQTS = 225$
- Correct answer without working award 3/4

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
10.			Ans: 25cm • ¹ correct fraction of circle • ² construct equation • ³ know how to solve equation • ⁴ solve equation and calculate length of the pendulum	4	• ¹ $\frac{65}{360}$ • ² e.g. $\frac{65}{360} \times \pi \times d = 28.4$ • ³ e.g. $d = \frac{28.4}{\frac{65}{360} \times \pi}$ • ⁴ 25

Notes:

1. Accept variations in π .

2. Accept 0.57 as evidence of $\frac{65}{360} \times \pi$ in awarding 2nd and 3rd marks

3. Disregard errors due to premature rounding provided there is evidence.

e.g. $d = \frac{28.4}{0.57} = 49.8 \rightarrow 24.9$ award 4/4

4. $\frac{65}{360} \times \pi \times r^2 = 28.4 \rightarrow 7.07 \dots, 7.1$ or 7 award 3/4 ✓ x ✓ ✓

5. For the award of the 4th mark, the calculation must include 28.4, a fraction (e.g. $\frac{65}{360}$ or 0.18...) and a division by π

6. Correct answer without working award 0/4

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
11.			Ans: 1039.2 cm² • ¹ correct angle • ² correct substitution into area of triangle formula • ³ know how to find area of hexagon • ⁴ correct calculation and correct units	4	• ¹ 60 • ² $\frac{1}{2} \times 20 \times 20 \times \sin 60$ • ³ $\left(\frac{1}{2} \times 20 \times 20 \times \sin 60 \right) \times 6$ • ⁴ 1039.2 cm ²

Notes:

1. Correct units must be given in the **final answer** for the award of the 4th mark.

2. Disregard errors due to premature rounding provided there is evidence.

e.g. $\sin 60 = 0.87 \Rightarrow \left(\frac{1}{2} \times 20 \times 20 \times 0.87 \right) \times 6 = 1044 \text{ cm}^2$ award 4/4

3. Some common answers:

(a) $\left(\frac{1}{2} \times 40 \times 40 \times \sin 60 \right) \times 6 = 4156.9 \text{ cm}^2$ award 3/4 ✓ x ✓ ✓

(b) $\frac{1}{2} \times 40 \times 40 \times \sin 60 = 692.8 \text{ cm}^2$ award 1/4 ✓ x x x

(c) $\left(\frac{1}{2} \times 20 \times 20 \right) \times 6 = 1200 \text{ cm}^2$ award 1/4 x x ✓ x

4. Use of GRAD or RAD (working must be shown)

(a) For 970.8 cm² [uses GRAD] award 4/4

(b) For -365.8 cm² or 365.8 cm² [uses RAD] award 3/4

5. Correct answer without working award 4/4

6. Alternative strategy (using $\frac{1}{2}bh$ to find area of triangle).

Award the marks as follows:

•¹ correct length of side of hexagon

•¹ 20

•² correct substitution into area of triangle formula

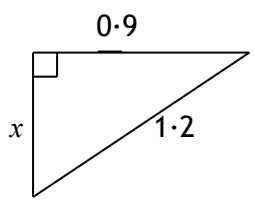
•² $\frac{1}{2} \times 20 \times \sqrt{20^2 - 10^2}$

•³ know how to find area of hexagon

•³ $\left(\frac{1}{2} \times 20 \times \sqrt{20^2 - 10^2} \right) \times 6$

•⁴ correct calculation and correct units

•⁴ 1039.2 cm²

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
12.			Ans: 1.99 metres •¹ marshal facts and recognize right-angle •² correct Pythagoras statement •³ correct calculation of x •⁴ find depth of milk	4	•¹  •² $x^2 = 1.2^2 - 0.9^2 (= 0.63)$ •³ 0.79..... •⁴ 1×99

Notes:

1. $x = 0.8$ and depth = 2 are acceptable in awarding the third and fourth marks
2. The final mark is for adding 1.2 to a value which has been calculated
3. In the absence of a diagram accept $x^2 = 1.2^2 - 0.9^2$ as evidence for the award of the first 2 marks
4. For $x^2 = 1.2^2 + 0.9^2 \rightarrow x = 1.5 \rightarrow \text{depth} = 2.7$
 - (a) with correct diagram award 3/4 ✓x✓✓
 - (b) without diagram award 2/4 xx✓✓
5. Where a candidate assumes angle MLO = angle OML = 45°, only the 1st and 4th marks are available
6. For an answer of 1.99 without working award 0/4

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
13.			Ans: 23.8 kilometres • ¹ calculate the size of angle PQR • ² correct substitution into sine rule • ³ know how to solve equation • ⁴ calculate PR correctly	4	• ¹ 52 • ² $\frac{q}{\sin 52} = \frac{25}{\sin 56}$ • ³ $q = \frac{25 \sin 52}{\sin 56}$ • ⁴ 23.8

Notes:

- Disregard errors due to premature rounding provided there is evidence
- Where incorrect sizes are used for angles, marks 3 and 4 are still available for rearranging and processing a sine rule calculation

e.g. $\frac{25}{\sin 160} = \frac{q}{\sin 128} \rightarrow q = 57.6$ award 2/4 x x ✓ ✓

- For a correct answer without working award 0/4

4. For $\frac{q}{52} = \frac{25}{56} \rightarrow q = 23.2 \dots$ award 1/4 ✓ x x x

- Use of GRAD or RAD (working must be shown)

(a) For 23.7 [uses GRAD]

award 4/4

(b) For -47.3 or 47.3 [uses RAD]

award 3/4

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
14.	(a)	(i)	Ans: $2x+13$ • ¹ correct expression	1	• ¹ $2x+13$
		(ii)	Ans: $4x^2 + 44x + 117 = 270$ $\Rightarrow 4x^2 + 44x - 153 = 0$ • ¹ find expression for area of card and expand pair of brackets • ² construct equation and rearrange into required form	2	• ¹ $(2x+13)(2x+9) = 4x^2 + 44x + 117$ • ² $4x^2 + 44x + 117 = 270$ $\Rightarrow 4x^2 + 44x - 153 = 0$

Notes:

1. If solution to (a)(ii) appears in (b) then both marks are available

	(b)		Ans: $x = 2.8$ cm • ¹ correct substitution into quadratic formula • ² evaluate discriminant • ³ solve for x • ⁴ select positive value of x , correctly stated to 1 decimal place	4	• ¹ $x = \frac{-44 \pm \sqrt{44^2 - 4 \times 4 \times (-153)}}{2 \times 4}$ • ² $x = \frac{-44 \pm \sqrt{4384}}{2 \times 4}$ (stated or implied by • ³) • ³ $x = 2.77... \text{ and } -13.77....$ • ⁴ $x = 2.8$
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Notes:

1. If solution to (b) appears in a(ii) then all four marks are available.

However, if a different value for x is stated in (b) then the fourth mark is not available.
(General Marking Principle (i) should not be applied in this special case.)

2. Where $b^2 - 4ac$ is calculated incorrectly, the third and fourth marks are only available if
 $b^2 - 4ac > 0$.

3. Where a, b and c are all positive the second mark is not available.

4. Correct answer without working award 0/4

[END OF MARKING INSTRUCTIONS]

General Marking Principles for National 5 Mathematics

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this Paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

- (a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.
- (b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.
- (c) If a specific candidate response does not seem to be covered by either the principles or detailed Marking Instructions, and you are uncertain how to assess it, you must seek guidance from your Team Leader.
- (d) Credit must be assigned in accordance with the specific assessment guidelines.
- (e) Candidates may use any mathematically correct method to answer questions except in cases where a particular method is specified or excluded.
- (f) Working subsequent to an error must be followed through, with possible credit for the subsequent working, provided that the level of difficulty involved is approximately similar. Where, subsequent to an error, the working is easier, candidates lose the opportunity to gain credit.
- (g) Where transcription errors occur, candidates would normally lose the opportunity to gain a processing mark.
- (h) Scored out or erased working which has not been replaced should be marked where still legible. However, if the scored out or erased working has been replaced, only the work which has not been scored out should be judged.
- (i) Where a candidate has made multiple attempts, mark all attempts and award the lowest mark.
- (j) Unless specifically mentioned in the specific assessment guidelines, do not penalise:
 - Working subsequent to a correct answer
 - Correct working in the wrong part of a question
 - Legitimate variations in solutions
 - Bad form
 - Repeated error within a question